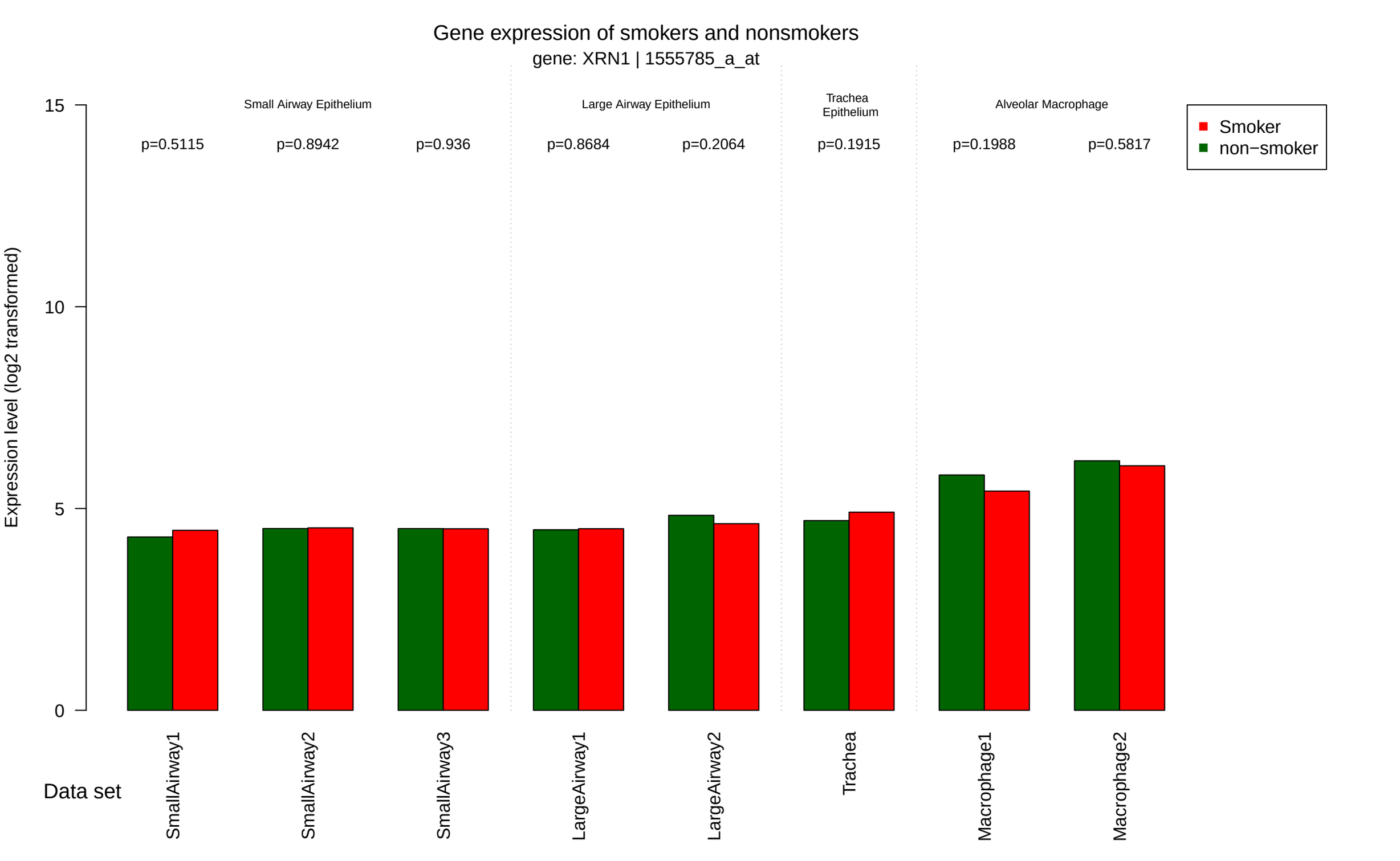
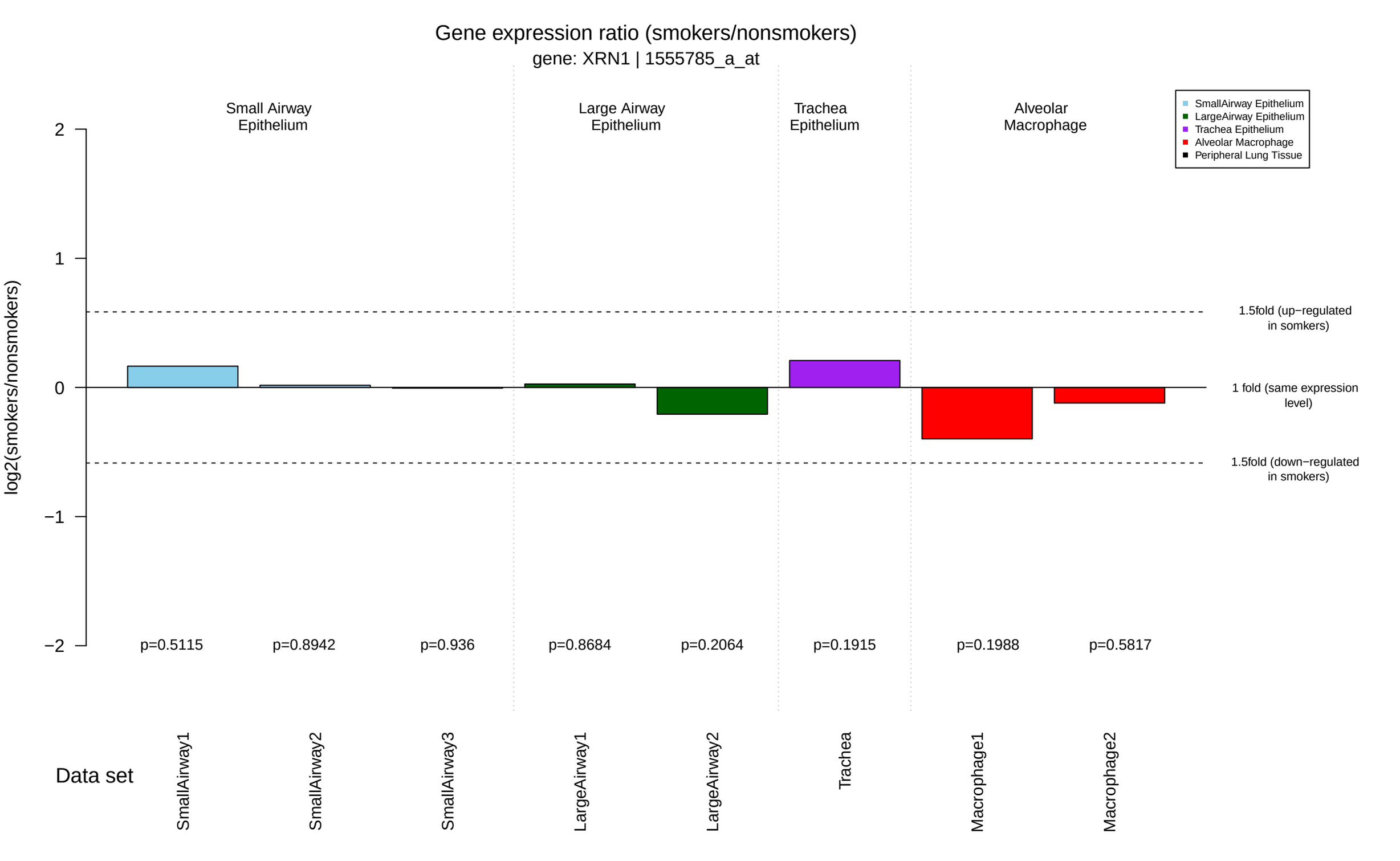
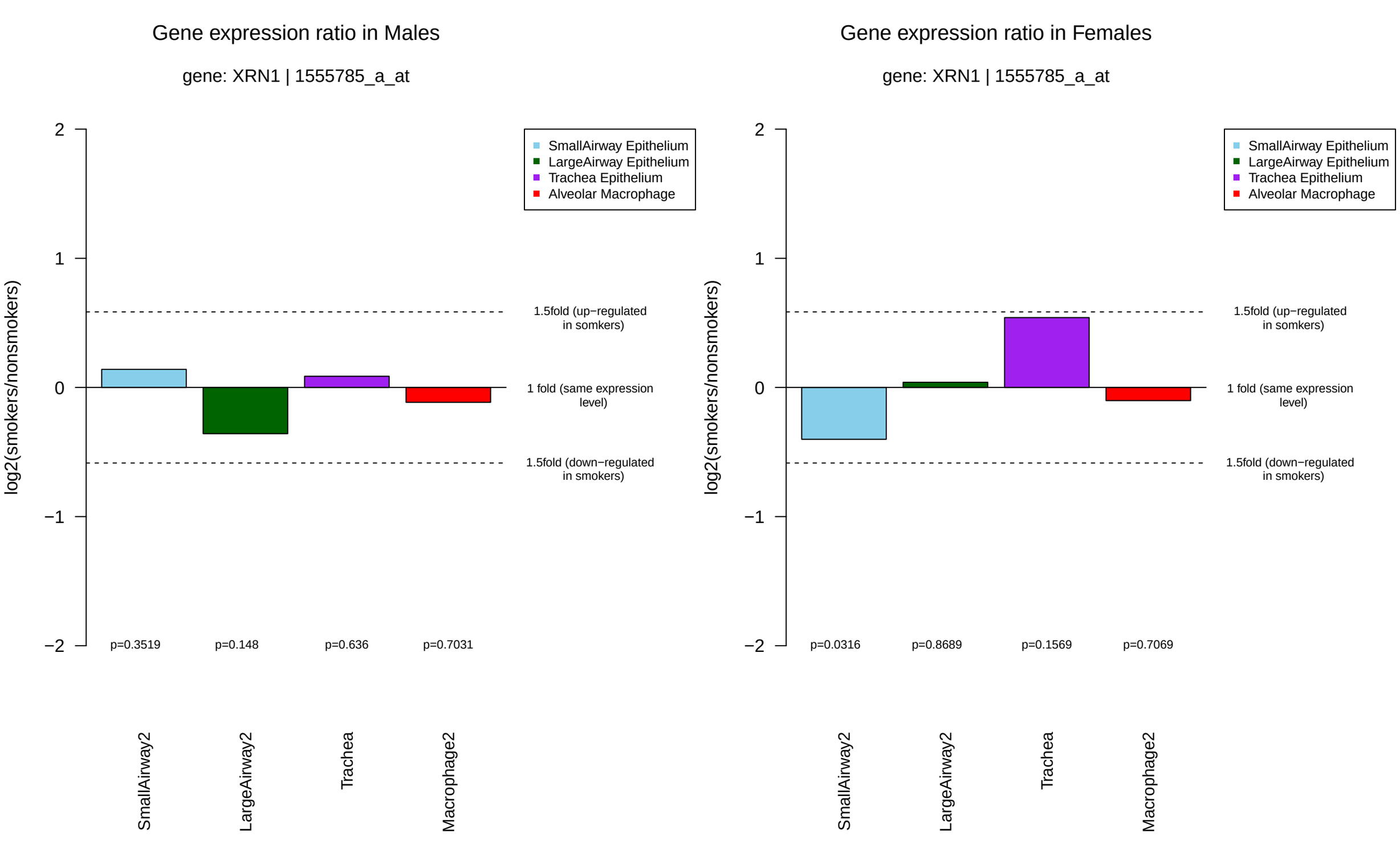
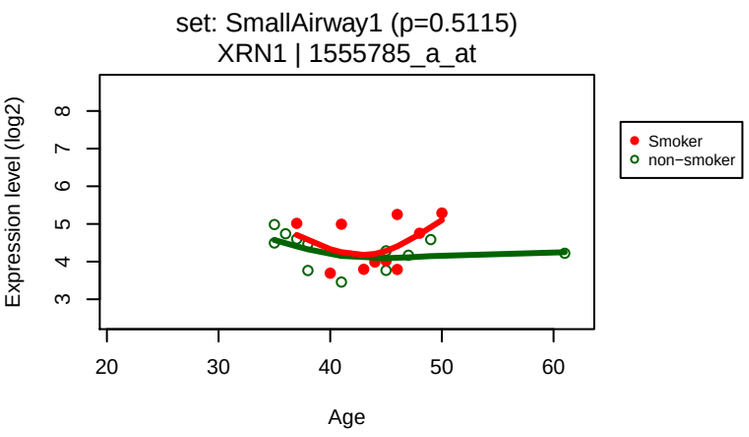
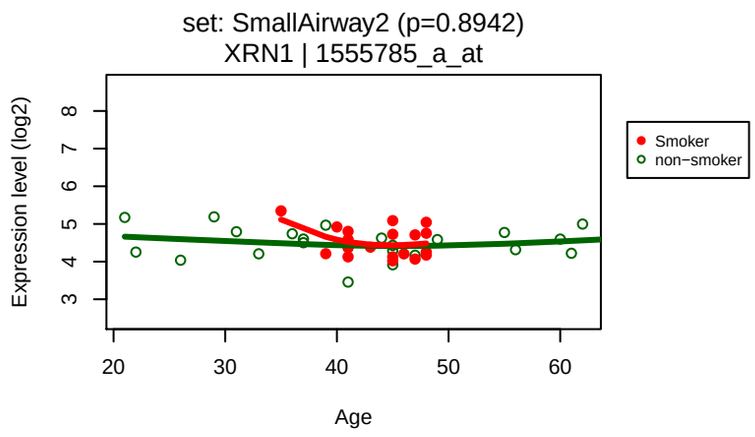
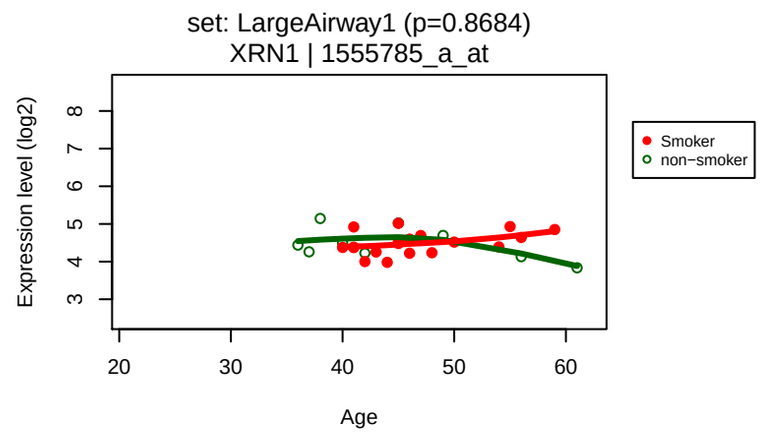

Gene expression of smokers and nonsmokers
gene: XRN1 | 1555785_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5115
p=0.8942
p=0.936
p=0.8684
p=0.2064
p=0.1915
p=0.1988
p=0.5817
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

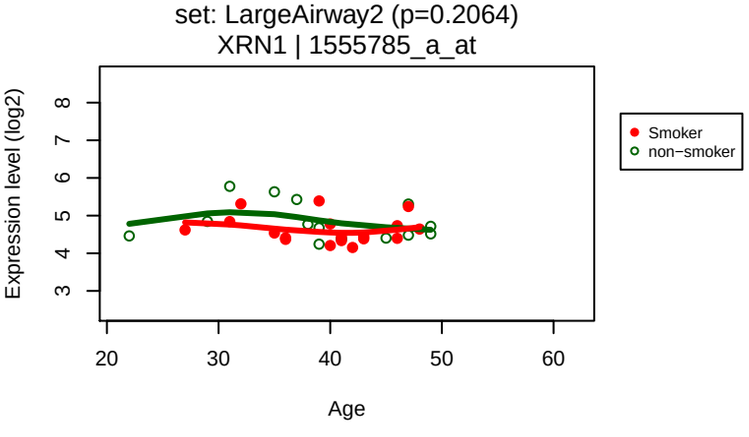

Gene expression ratio (smokers/nonsmokers)
gene: XRN1 | 1555785_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5115
p=0.8942
p=0.936
p=0.8684
p=0.2064
p=0.1915
p=0.1988
p=0.5817
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

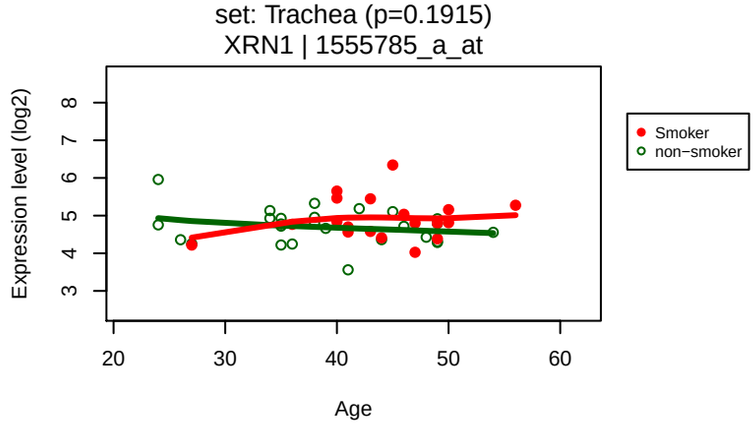

Gene expression ratio in Males
gene: XRN1 | 1555785_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3519
p=0.148
p=0.636
p=0.7031
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: XRN1 | 1555785_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0316
p=0.8689
p=0.1569
p=0.7069
SmallAirway2
LargeAirway2
Trachea
Macrophage2

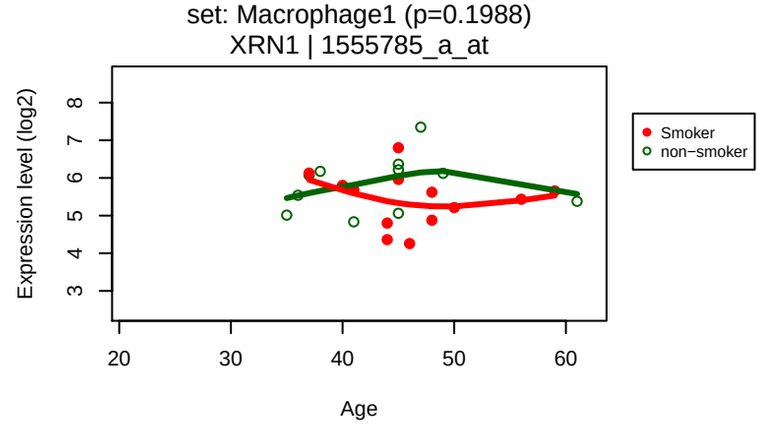

set: SmallAirway1 (p=0.5115)
XRN1 | 1555785_a_at
Expression level (log2)
Age
Smoker
non−smoker

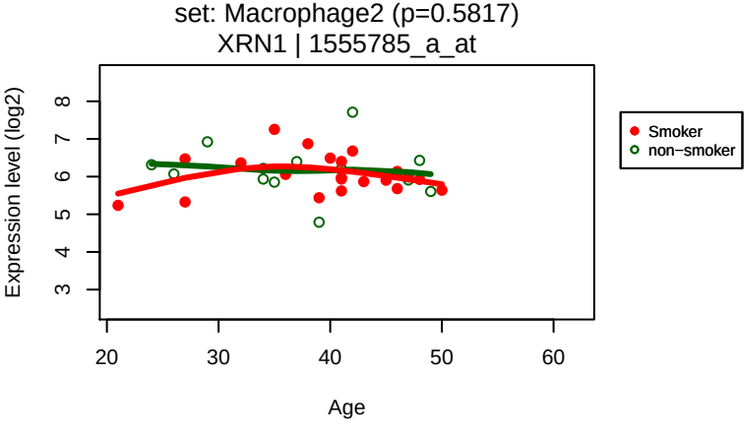

set: SmallAirway2 (p=0.8942)
XRN1 | 1555785_a_at
Expression level (log2)
Age
Smoker
non−smoker

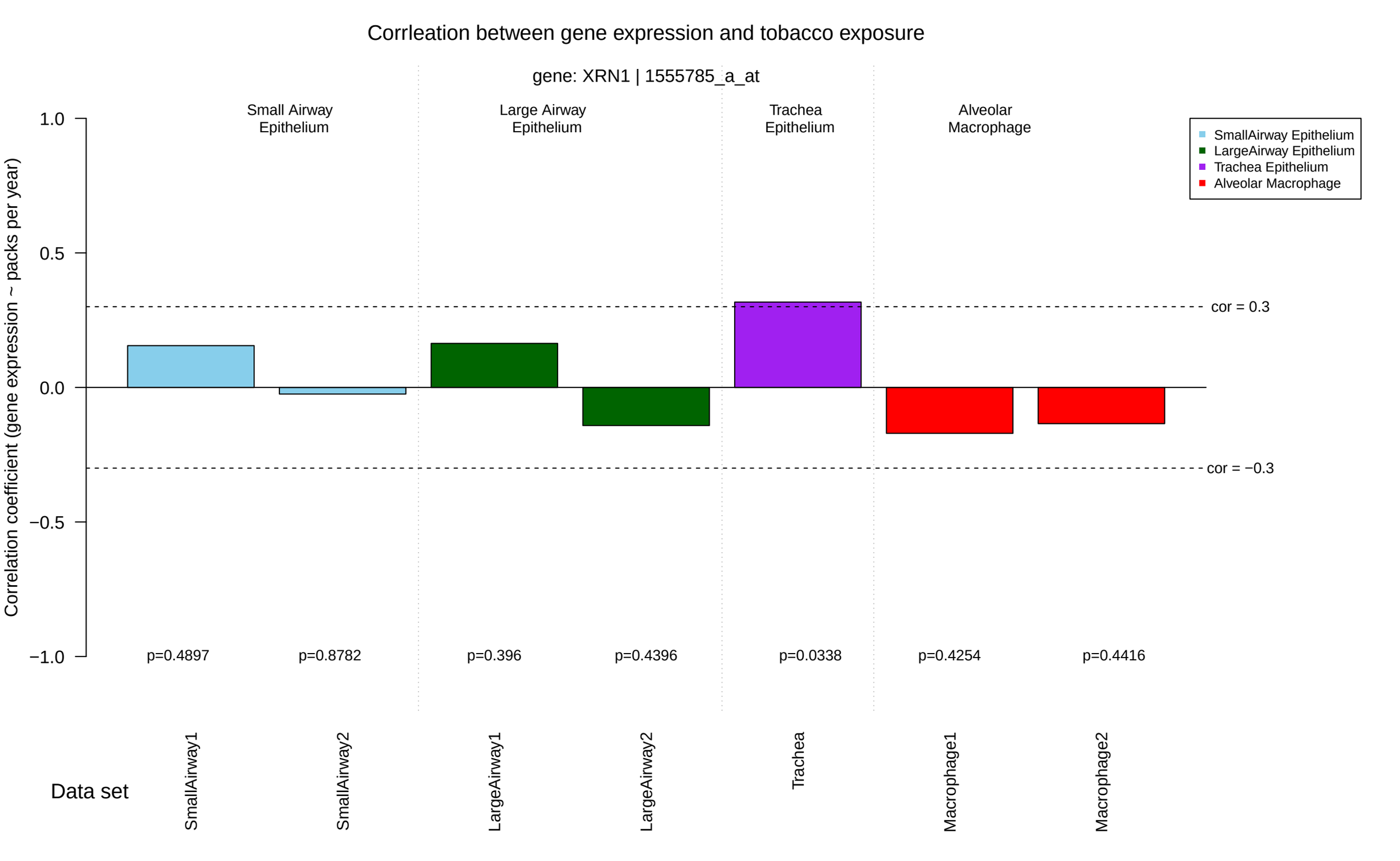

set: LargeAirway1 (p=0.8684)
XRN1 | 1555785_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2064)
XRN1 | 1555785_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1915)
XRN1 | 1555785_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1988)
XRN1 | 1555785_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5817)
XRN1 | 1555785_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: XRN1 | 1555785_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4897
p=0.8782
p=0.396
p=0.4396
p=0.0338
p=0.4254
p=0.4416
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2